Cause Entity (C)

Canonical: PmrB

Sensor kinase • p. 2

DIRECT_CAUSE (+)

conf = 0.95 • Forward • Results

Effect Entity (E)

Canonical: PmrA

Response reg. • p. 3

Evidence Sentence (S_causal) **100% PROVENANCE GROUNDED**

"Acid pH activates PmrA/PmrB by promoting PmrB-mediated phosphorylation of PmrA."

Doc: PMC1804205 • Section: Results • Page: 2 • Offsets: [1420–1512]

1. Direct Assertion

Single sentence boundary

- Co-occurs in single sentence
- Trigger: activates, induces
- Exact offsets [1420–1512], p. 2
- Precision 0.90–0.98 on gold

2. Context-Resolved

Cross-sentence bridge

- Aligned across [S_k, S_k+1] spans
- Demonstrative anaphora resolution
- Dual-sentence provenance binding
- Acronym & synonym mapping

3. Transitive Link

Multi-page graph traversal

- Path: A (p.2) → B → C (p.8)
- Page $\Delta p \geq 1$, hops ≤ 2
- Pruned by domain invariants
- Calibrated score compounding